Name: _____ **Period:** _____ **Date:** _____

Topic 2.2 Guided Notes: Thinking, Problem-Solving, and Decisions

Unit 2: Cognition | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic2.2

Word Bank

Convergent Thinking	*Creativity*	*Assimilation*	*Mental Set*
Divergent Thinking	*Sunk-Cost Fallacy*	*Gambler's Fallacy*	*Representativeness Heuristic*
Framing	*Prototype*	*Heuristics*	*Executive Functions*
Accommodation	*Critical Thinking*	*Schemas*	*Functional Fixedness*
Concepts	*Algorithms*	*Priming*	*Availability Heuristic*

1. The Building Blocks of Thought

To make sense of the world, we have to organize it. We do this by grouping similar objects, ideas, or events into mental categories called _____. If I ask you to picture a "bird," you probably won't imagine a penguin or an ostrich; you'll likely picture a robin or a sparrow. That is because the robin is your _____ — the most typical, perfect, or best example of a concept.

As we grow, we build vast mental frameworks called _____ based on our experiences. When we encounter something new, our brain tries to fit it into an existing schema — a process called _____. For example, a toddler who has only ever seen dogs might point to a cow and say "Doggie!" because it has four legs and a tail. But eventually, the child learns that cows moo and dogs bark. They must modify their existing schema to create a brand new category for cows. This cognitive adjustment is called _____.

Apply It: Which Building Block?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	When asked to picture "furniture," most people picture a chair or couch rather than a beanbag or ottoman.	
2	A child who knows about apples immediately calls a new red pear a piece of fruit without needing to change any of her thinking.	
3	After being stung, a child who always thought bees were friendly like butterflies creates a brand new mental category for dangerous insects.	

2. Problem Solving: The Slow Way vs. The Fast Way

When faced with a problem, our brain's _____ kick in. These are the higher-level cognitive processes that allow us to generate ideas, organize, plan, and carry out goal-directed behaviors — the same mental toolkit responsible for _____, since evaluating evidence and weighing options well is really just executive functions in action.

- _____: step-by-step, logical procedures that guarantee a solution if followed correctly. Imagine forgetting the 3-digit combination to your padlock — trying 0-0-0, then 0-0-1, all the way to 9-9-9 will eventually open the lock, but it will take hours.
- _____: simple, efficient "rules of thumb" or mental shortcuts. Instead of trying every number on the lock, this approach would be guessing your birthday or lucky number — incredibly fast, but it doesn't guarantee a correct answer.

Apply It: Algorithm or Heuristic?

#	Scenario	Concept
1	A baker follows the exact same step-by-step recipe measurements every time to guarantee a perfect cake.	
2	Rather than reading every restaurant review individually, a hungry traveler just picks the restaurant with the most stars.	

📺 Video Notes: Cognition — How Your Mind Can Amaze and Betray You (Crash Course Psychology #15)

This video reviews concepts, prototypes, and problem-solving, and introduces a few terms not covered on the page (insight, confirmation bias, belief perseverance). Answer the questions below as you watch.

- The video describes three problem-solving approaches: trial and error, algorithms, and heuristics. How does it describe each, and what Sriracha example does it use to compare algorithms and heuristics?

- What is insight, according to the video, and what did neuroimaging show in the CRAB, PINE, SAUCE experiment while subjects worked on the problem and at the moment of their “aha!”?

- What are confirmation bias and belief perseverance, according to the video, how are they connected, and what example does the video give of belief perseverance?

- What are concepts and prototypes, according to the video, and how can they lead to prejudice? Use the video's bird and doctor examples.

- What is the availability heuristic, according to the video, who researched it, and how does the casino example illustrate it?

3. When Shortcuts Betray Us (Cognitive Biases)

Because we rely so heavily on heuristics, we are vulnerable to predictable errors in judgment:

- _____: estimating the likelihood of an event based on how easily a vivid example comes to mind. People are often terrified of shark attacks but don't think twice about driving on the highway, simply because shark attacks are vividly covered in the news.
- _____: judging a situation based on how well it matches your prototype or stereotypes. Seeing someone who is quiet, loves reading, and wears glasses might lead you to assume they're a librarian rather than a salesperson.
- _____: the mistaken belief that past random events influence future independent events. If a coin lands on heads five times in a row, someone might bet heavily on tails, thinking it is "due" — in reality, the odds are still 50/50.
- _____: the tendency to continue an endeavor because you have already invested time or money into it, even when the future costs outweigh the benefits (e.g., staying in a movie theater to watch a terrible movie just because you paid $15 for the ticket).

Apply It: Which Cognitive Bias?

#	Scenario	Bias
1	After hearing several dramatic news stories about lottery winners, a man believes his odds of winning are much higher than they truly are.	
2	A woman assumes a man wearing a suit and carrying a briefcase must be a lawyer rather than a teacher, simply because he matches her mental image of a lawyer.	
3	After a roulette wheel lands on black six times in a row, a gambler is now certain that red is "due" to hit next.	
4	A company continues funding a clearly failing project only because of how much money has already been spent developing it.	

4. Thinking Outside the Box (and What Traps Us)

_____ is the ability to produce original and valuable ideas. It requires _____ — exploring many different possible solutions to a problem (like brainstorming 50 different uses for a brick). Schools, however, often test _____, which demands narrowing down possibilities to find the single correct answer (like a multiple-choice test).

- _____: the tendency to approach a problem using the exact same strategy that worked in the past, even if it isn't helpful for the current situation.
- _____: a cognitive bias that limits a person to using an object only in its traditional way. If you need to tighten a screw but don't have a screwdriver, this bias might prevent you from realizing that the dime in your pocket would work perfectly.

Apply It: Which Concept?

#	Scenario	Concept
1	A design team is asked to come up with as many different possible uses for a paperclip as they can, with no wrong answers.	
2	A student takes a standardized test that requires arriving at the one single correct answer for each question.	
3	A cook who always seasons every dish the exact same way struggles to adjust when a recipe calls for a completely different flavor profile.	
4	Needing a makeshift funnel, a camper doesn't realize that a water bottle cut in half could work, since he only thinks of it as something to drink from.	

5. The Invisible Hands of Context

Our decisions are constantly being manipulated by external circumstances we aren't even aware of:

- _____: the unconscious activation of specific associations in memory. If you walk past a bakery smelling fresh bread, you might be more likely to buy a dessert at the restaurant next door without realizing why.
- _____: how information is presented significantly impacts our decisions. A surgeon telling a patient a procedure has a "90% survival rate" sounds much more appealing than saying it has a "10% mortality rate," even though the underlying data is identical.

Apply It: Priming or Framing?

#	Scenario	Concept
1	After watching a scary movie about sharks, a swimmer feels unusually nervous wading into calm ocean water the next day.	
2	A store advertises a jacket as "75% off" rather than saying customers "still pay 25% of the original price," even though the deal is identical.	

6. Don't Trip Up! (Common Misconceptions)

Think of the "ss" in aSSimilation as meaning "Same Schema." Think of the "cc" in aCCommodation as meaning "Change Concept" or "Create Category!"

_____ is about Memory — did you just see a vivid news story about a plane crash?
_____ is about Stereotypes — does this person match my mental prototype of a librarian?

7. Quick Check: Review Questions

Circle the best answer for each question below.

1. When asked to picture a "dog," most people immediately imagine an animal like a Labrador rather than a rare, unusual breed. The Labrador represents the:

- **A.** Algorithm
- **B.** Concept
- **C.** Prototype
- **D.** Schema

2. A toddler who has only ever seen dogs points at a cow and confidently says "Doggie!" because it also has four legs and a tail. After his parents explain that cows moo instead of bark, he creates an entirely new mental category just for cows. This adjustment is called:

- **A.** Accommodation
- **B.** Convergent Thinking
- **C.** Assimilation
- **D.** Priming

3. Instead of trying every possible number on his lock, a man decides to first guess his own birthday, reasoning that it's a quick, likely guess even though it isn't guaranteed to be correct. This fast, efficient "rule of thumb" strategy is called a(n):

- **A.** Executive Function
- **B.** Heuristic
- **C.** Algorithm
- **D.** Mental Set

4. Meeting a new coworker who is quiet, wears glasses, and enjoys reading during breaks, a man immediately assumes she must work in the IT department rather than in sales, simply because she matches his mental image of a typical IT employee. This judgment is best explained by the:

- **A.** Availability Heuristic
- **B.** Gambler's Fallacy
- **C.** Functional Fixedness
- **D.** Representativeness Heuristic

5. Despite finding a movie boring and poorly made within the first 20 minutes, a couple decides to sit through the entire painful two-hour film, reasoning that they already paid $15 each for their tickets and don't want that money to go to waste. This decision reflects the:

- **A.** Mental Set
- **B.** Sunk-Cost Fallacy
- **C.** Functional Fixedness
- **D.** Gambler's Fallacy

6. During a brainstorming session, a team is asked to generate as many different possible uses for an ordinary brick as they can think of, with no single right answer being sought. This kind of open-ended thinking is called:

- **A.** Mental Set
- **B.** Functional Fixedness
- **C.** Divergent Thinking
- **D.** Convergent Thinking

7. Needing to prop open a heavy door but lacking an official doorstop, a man struggles for several minutes without realizing that the thick hardcover book sitting right next to him on the shelf would work perfectly. His inability to see this book as anything other than something to read is an example of:

- **A.** Functional Fixedness
- **B.** Mental Set
- **C.** Gambler's Fallacy
- **D.** Sunk-Cost Fallacy

✅ Check Your Progress

Done with your notes? Head over to the Quiz to test what you've learned:
https://appsychreviewhub.com/quiz

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